Initial Setup

- Verify that Dockerfile and docker-compose.yml are in your "C:\Users\...\IEM4723_The_Detectives\Code\" file
    - If they are not, pull the branch TY-Gurobi-Container and make sure your computer is using that branch
- To get your gurobi license, do the following:
    - Log in or create an account using your student email: portal.gurobi.com/iam/login/
    - Click the Licenses tab on the left of the screen
    - If you have a license of type WLS Compute Server, click the button in its row
        - If not, click request on the left and then Generate now under the WLS option
    - Download the WLS license. You will get a .lic file. Make sure it's named gurobi.lic, then move it into the "C:\Users\...\IEM4723_The_Detectives\Code\" file on your computer
- Optionally, in the docker terminal, navigate to "C:\Users\...\docker_tutorial\mysql-dev\" using the cd command and run "docker compose stop"
    - Then navigate to "C:\Users\...\docker_tutorial\redis-dev" and run "docker compose stop"
    - These will pause the class containers we've been running to save memory on your device. To restart them, navigate back to their folders using cd commands and run "docker compose start"

Docker Container

All steps are to be done in the Docker terminal

- Navigate to the "C:\Users\...\IEM4723_The_Detectives\Code\" file on your computer using a cd command
- Create the image and run it using "docker compose up -d --build"
- Verify the license was mounted using "docker compose exec jupyter cat /opt/gurobi/gurobi.lic"
- Verify the data was mounted using "docker compose exec jupyter ls /home/jovyan/data"

Code Setup

- Find the file ContainerSetup.ipynb
- Open it in JupyterLab through localhost:4444
- Run all cells, verifying includes the output below:
    - Cell 1: Academic license 2772870 - for non-commercial use only - registered to ___@okstate.edu, then an output for the example optimization problem, which should have objective value 3.200000000e+00
    - Cell 2: b'hello'
    - Cell 3: (('IEM4723_The_Detectives',), ('iem4723_mysql_tutorial')...)
    - Cell 4: No output
- Open localhost:2222 to check that IEM4723_The_Detectives is present in phpMyAdmin

- Once you have verified all connections work, you may use the example connection instances in ContainerSetup.ipynb in your own .ipynb files. Please do not change the file itself.